

Retrieval-Augmented Generation (RAG) is a technique that combines a retrieval system with a generative language model. The retrieval system finds relevant documents for a given question, and the language model uses those documents to generate a more accurate, grounded answer.

RAG was popularised in 2020 by Facebook AI Research. The key insight is that large language models, while powerful, are bounded by what they saw during training. RAG lets you extend their knowledge to any corpus you choose, at inference time, without retraining.

The core components of a RAG system are: a document loader (which reads PDFs, HTML, or other formats), a chunker (which splits long documents into smaller pieces), an embedder (which turns each chunk into a vector), a vector store (which indexes the vectors for fast similarity search), and a language model (which generates the final answer using the retrieved chunks as context).

The chunker is often overlooked, but its parameters significantly affect answer quality. Common defaults are 200-800 tokens per chunk with 10-20 percent overlap between adjacent chunks.

Vector stores use approximate nearest neighbour algorithms to find similar vectors quickly. Common open source vector databases include ChromaDB, FAISS, Qdrant, and Milvus. Production systems often use managed services like Pinecone or Weaviate Cloud.

The choice of embedding model matters a lot. Strong general-purpose embedding models include `openai-text-embedding-3-small` (open source, runs locally), `text-embedding-3-small` from OpenAI, and `voyage-3` from Voyage AI. `Sentence-transformers` is a popular library that hosts many open embedding models that can run on CPU or GPU.